(1) Find the domain and the range of $f(x)=\sqrt{9-x^2}$.

Solution : $f(x) = \sqrt{9-x^2}$

$f(x)$ is defined,

if $9-x^2 \geq 0$

i.e. $9 \geq x^2$,

i.e if $x^2 \leq 9$

i.e. if $-3 \leq x \leq 3$

$\therefore$ Domain $= \{x \mid x \in R, -3 \leq x \leq 3\}$

$= [-3, 3]$

when $-3 \leq x \leq 3$, $0 \leq x^2 \leq 9$

$\therefore 0 \geq -x^2 \geq -9$

$\therefore 0+9 \geq 9-x^2 \geq 9-9$

$\therefore 9 \geq 9-x^2 \geq 0$

$\therefore 0 \leq 9-x^2 \leq 9$

$\therefore 0 \leq \sqrt{9-x^2} \leq 3$

$\therefore 0 \leq f(x) \leq 3$

$\therefore 0 \leq y \leq 3$,

where $y = f(x)$

Range $= \{y \mid y \in R, 0 \leq y \leq 3\}$

$= [0, 3]$

$\therefore$ Domain $= [-3, 3]$

Range $= [0, 3]$

2) If $f: A \to B$ and $g: B \to C$ are one-one functions then show that $g \circ f: A \to C$ is also one-one.

Solution : Here $f: A \to B$ and $g: B \to C$ are one-one function.

Let $x_1, x_2 \in A$

Now $(g \circ f)(x_1) = (g \circ f)(x_2)$

$\Rightarrow g(f(x_1)) = g(f(x_2))$

$\Rightarrow f(x_1) = f(x_2)$ ($\because g$ is one-one function)

$\Rightarrow x_1 = x_2$ ($\because f$ is one-one function)

$\therefore$ For $(g \circ f)(x_1) = (g \circ f)(x_2)$

We have $x_1 = x_2 \ \forall \ x_1, x_2 \in A$

$\therefore g \circ f$ is one-one function.

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3) Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$ find the value of $\log\sqrt{96}$.

Solution : $\log\sqrt{96} = \log(96)^{1/2}$

$= \frac{1}{2}\log(96)$

$= \frac{1}{2}\log(32\times3)$

$= \frac{1}{2}\log(2^5\times3)$

$= \frac{1}{2}[\log 2^5 + \log 3]$

$= \frac{1}{2}[5\log 2 + \log 3]$

$= \frac{1}{2}[5\times0.3010 + 0.4771]$

$= \frac{1}{2}[1.5050 + 0.4771]$

$= \frac{1}{2}[1.9821]$

$= 0.99105$

Theory : Vertical shift : For c > 0, $f(x)$ + c shifts graph of $f(x)$ to c units upward, and $f(x)$ - c shifts graph of $f(x)$ to c units downward.

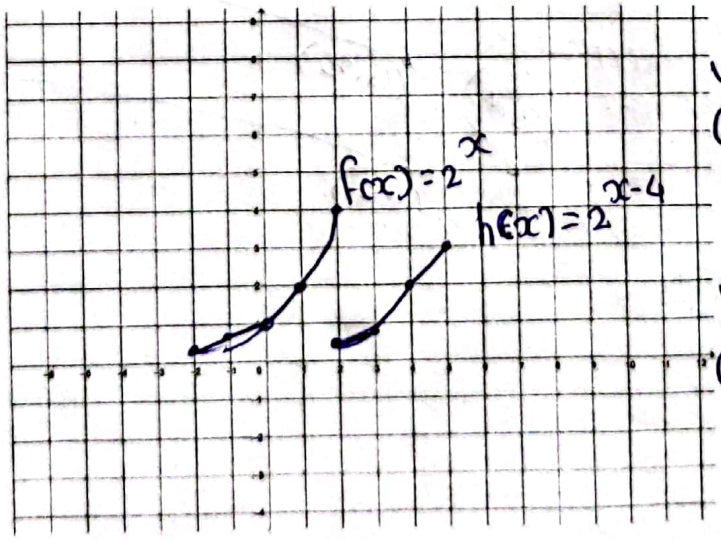
Horizontal shift : For c > 0, $f(x + c)$ shifts graph of $f(x)$ to c units to left, and $f(x - c)$ shifts graph of $f(x)$ to c units to right.

Reflection about X -axis : $-f(x)$ reflects the graph about X - axis

Reflection about Y -axis : $f(-x)$ reflects the graph about Y - axis

Activity :

1) Draw the graph of $f(x) = 2^x$ and $h(x) = 2^{x-4}$ on the same graph paper.



x	0	1	2	-1	-2
$y=2^x$	1	2	4	1/2	1/4
(x,y)	(0,1)	(1,2)	(2,4)	(-1,1/2)	

x	2	3	4	5
$y=2^{x-4}$	1/4	1/2	1	2
(x,y)	(2,1/4)	(3,1/2)	(4,1)	(5,2)

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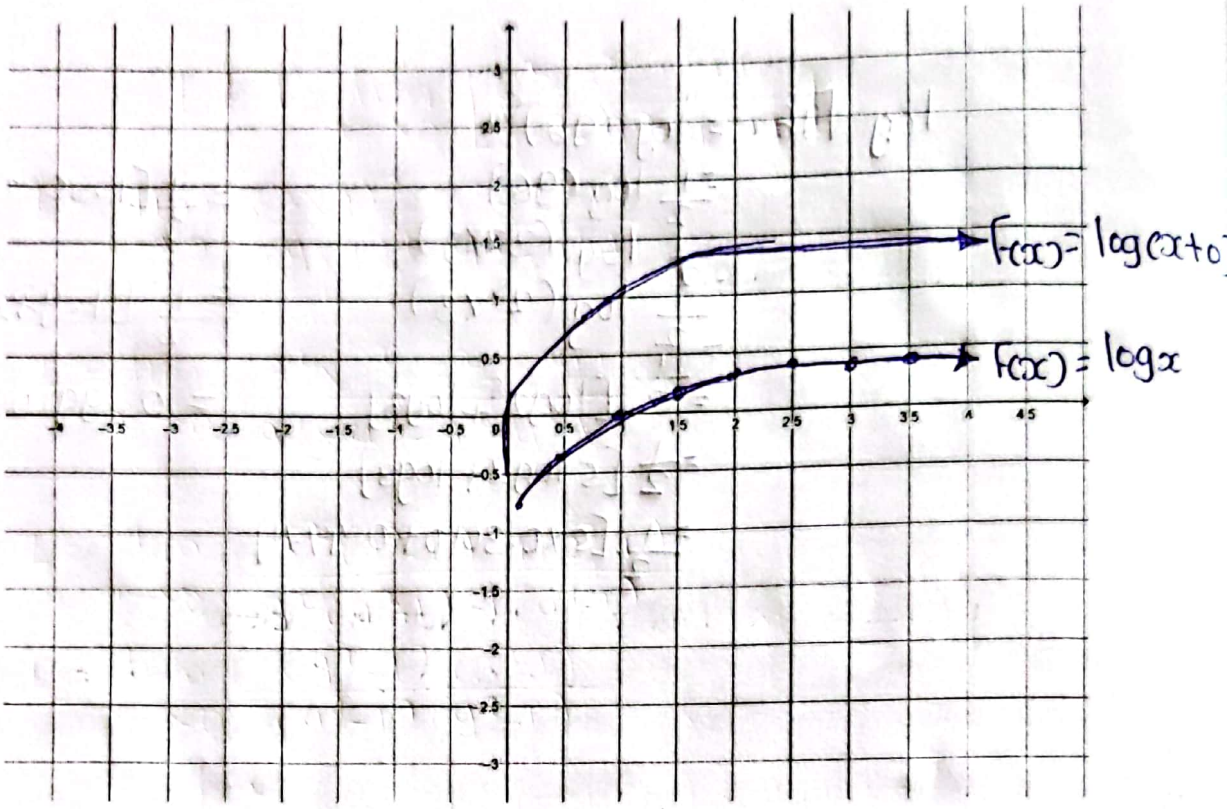
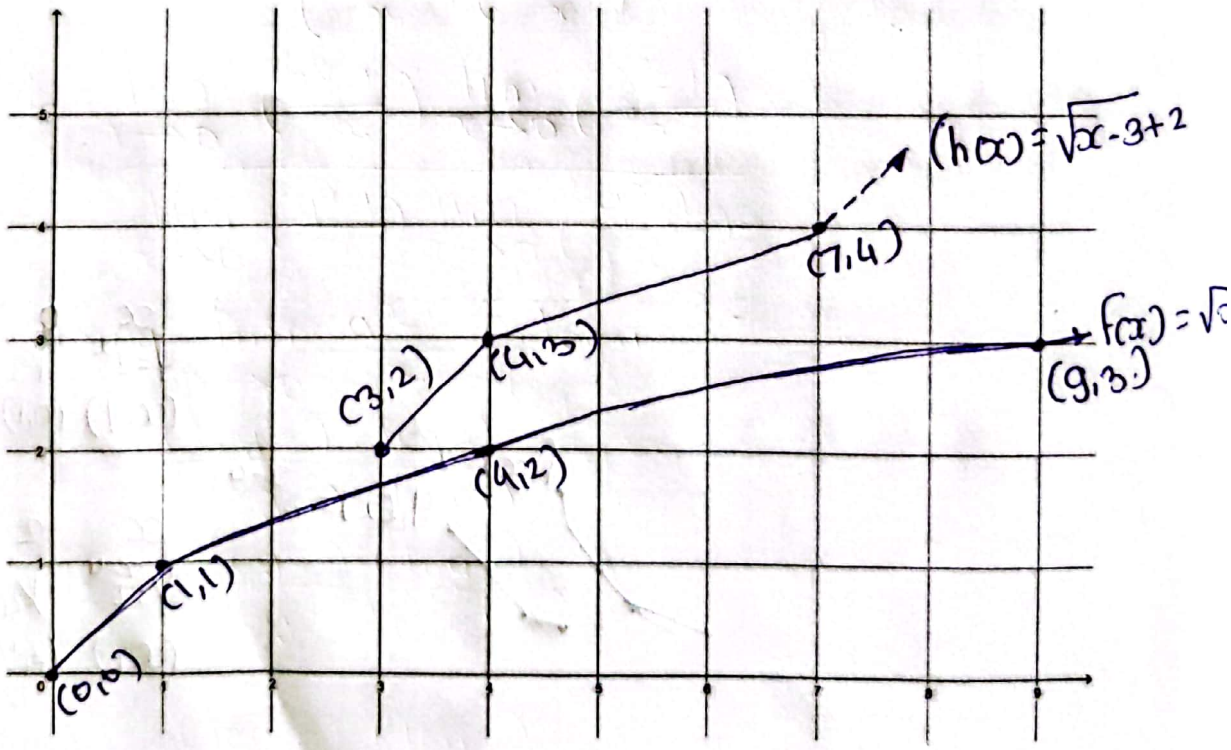
2) Draw the graph of $f(x) = \log x$ and $h(x) = \log(x + 1)$ on the same graph paper.



3) Draw the graph of $f(x) = \sqrt{x}$ and $h(x) = \sqrt{x-3}+2$ on the same graph paper.



Sign of Teacher :

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